



## CS23431 - OPERATING SYSTEM

### EXP 1: FCFS Using C

#### QUESTION :

Write a C program to implement the First-Come, First-Served (FCFS) CPU Scheduling Algorithm. The program should take the number of processes, their Arrival Times (AT), and Burst Times (BT) as input. Calculate and display the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process, along with the Average Turnaround Time and Average Waiting Time.

#### CODE:

```
#include <stdio.h>

int main()
{
    int n,i;
    float avg_tat=0,avg_wt=0;
    printf("Enter number of processes: \n");
    scanf("%d",&n);
    int at[n],bt[n],ct[n],tat[n],wt[n];
    for(i=0;i<n;i++)
    {
        printf("Enter Arrival Time and Burst Time for P%d: \n",i+1);
        scanf("%d %d",&at[i],&bt[i]);
    }
    ct[0]=at[0]+bt[0];
    for(i=1;i<n;i++)
    {
        if(ct[i-1] < at[i])
            ct[i]=at[i]+bt[i];
        else
            ct[i]=ct[i-1]+bt[i];
    }
    for(i=0;i<n;i++)
    {
        tat[i]=ct[i]-at[i];
        wt[i]=tat[i]-bt[i];
        avg_tat+=tat[i];
        avg_wt+=wt[i];
    }
    avg_tat/=n;
```



```
avg_wt/=n;
printf("\nAverage Turnaround Time: %.2f\n",avg_tat);
printf("Average Waiting Time: %.2f\n",avg_wt);
return 0;
}
```

## Output:

Custom Test

Success

Input:

Expected Output :

3  
0 2  
1 2  
2 3

Enter number of processes:  
Enter Arrival Time and Burst Time for P1:  
Enter Arrival Time and Burst Time for P2:  
Enter Arrival Time and Burst Time for P3:  
  
Average Turnaround Time: 3.33  
Average Waiting Time: 1.00

Output:

Enter number of processes:  
Enter Arrival Time and Burst Time for P1:  
Enter Arrival Time and Burst Time for P2:  
Enter Arrival Time and Burst Time for P3:  
  
Average Turnaround Time: 3.33  
Average Waiting Time: 1.00